

17 Modelling

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2026 objectives: 4.1

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CA1-P1 April 2005 Q4	11 marks
CA1-P1 April 2005 Q8	20 marks
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CA1-P1 | April 2005 | Q4

[Current topic](#) | [11 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 4** The financial regulations for insurance companies are being changed and as a result a company needs to introduce a stochastic asset model.

Describe the evaluation criteria for selecting a stochastic asset model. [11]

CA1-P1 | April 2005 | Q8

Current topic | 20 marks total | Original PDF p3

Related: Ch 19, Ch 25.

- 8** An investment company is considering selling a financial product in a developed overseas market. This will be its first international venture. The company has estimated the potential cashflows in today's monetary values under a number of different scenarios as follows.

Scenario	A	B	C
Year	\$000	\$000	\$000
1	-565	-565	-565
2	180	160	100
3	210	176	100
4	240	194	80
5	240	213	70
Probability of occurrence	30%	60%	10%

- (i) Explain how the real risk discount rate to be used when valuing this project should be chosen. [8]
- (ii) Calculate the net present value under each scenario and the expected net present value of the project. Use a real risk discount rate of 9% p.a. and assume that the cashflows occur midway through each year. State the formulae used. [6]
- (iii) Discuss the considerations which should be taken into account when deciding whether the project should proceed. [6]
- [Total 20]

CA1-P1 | April 2006 | Q9

Current topic | 21 marks total | Original PDF p3

Related: Ch 19, Ch 25.

- 9** A food retailer is considering introducing an organic food range. It is carrying out an initial appraisal of the proposed project.
- (i) Discuss the appraisal methods the food retailer should use to determine whether to pursue this project. [7]
 - (ii) Identify the major risks involved with the project and suggest how they might be mitigated. [6]
 - (iii) Describe how the discount rate to be used in evaluating this project should be chosen. [8]
- [Total 21]

CA1-P1 | September 2006 | Q7

Current topic | 22 marks total | Original PDF p3

Related: Ch 19.

- 7 A company is designing a stochastic asset model to be used for:
- evaluating investment management policies
 - calculating fair values; and
 - setting capital requirements
- (i) Describe the features required by such a stochastic asset model, in particular those that make it suitable for each of the required purposes. [12]
- (ii) Discuss the difficulties in designing one asset model for all these purposes using a single set of parameters. [10]
- [Total 22]

CA1-P1 | September 2007 | Q3

Current topic | 8 marks total | Original PDF p2

Related: Ch 36, Ch 19.

- 3** A life insurance company is setting up a new line of business.
- (i) Describe the modelling necessary to assess the capital required. [5]
 - (ii) List the principal experience assumptions that may be needed for the model in part (i). [3]
- [Total 8]

CA1-P1 | April 2010 | Q1

[Current topic](#) | [5 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 1** Outline the key requirements of a model that will be used to assess the variability of an outcome that depends on uncertain future events. [5]

CA1-P1 | April 2012 | Q4

Current topic | 12 marks total | Original PDF p3

Related: Ch 30.

- 4** (i) Outline why companies in the financial services sector have moved to using more complex models when designing and monitoring products. [3]

A large financial institution is about to launch an investment fund designed for experienced investors. The fund will invest directly in a large number of markets and investment classes.

The institution also intends to offer an insurance policy linked to the new fund. The insurance policy will provide protection against large falls in the value of an investor's holding in the new fund.

The institution will know what assets the fund will invest in. Hence the team responsible for designing and managing the fund believe that the insurance policy premiums can be invested in derivatives so as to hedge the risks faced by investors in the fund.

Therefore, by charging an appropriate premium, the design team says that there should be little risk to the institution in offering the insurance policy and guaranteed profits could be made.

The specific investments for the insurance policy will need to be determined by the design team using a complex model.

- (ii) Explain why the need to use a complex model may make it difficult for the directors of the institution to understand the actual financial risks associated with the investment of the insurance policy premium. [5]

- (iii) Discuss why the proposed hedging arrangement may not result in guaranteed profits for the institution. [4]

[Total 12]

CA1-P1 | April 2013 | Q2

[Current topic](#) | [8 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 2** Describe the requirements which should be satisfied when planning, designing and building a model. [8]

CA1-P1 | September 2013 | Q4

Current topic | 11 marks total | Original PDF p2

Related: Ch 36, Ch 28.

- 4** (i) List four ways in which scenarios could be derived for use in a capital model. [2]
- (ii) Outline the issues that an insurance company should consider when building a capital model. [9]
- [Total 11]

CA1-P1 | April 2015 | Q1

Current topic | 6 marks total | Original PDF p2

Related: Ch 3.

- 1** A country has recently set up a new regulator of occupational benefit schemes, following public concern over some cases where occupational benefit schemes have failed to provide benefits that had been promised to employees.

The regulator will have a wide range of statutory powers. It has announced that it will require all schemes to supply information about their assets and liabilities. The regulator will use this information to compare funding positions for all the different schemes and decide what (if any) further action to take in each case.

Outline the advantages and disadvantages to the regulator of using a standard model in order to compare schemes for this purpose. [6]

CA1-P1 | September 2015 | Q5

[Current topic](#) | [16 marks total](#) | [Original PDF p3](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 5** (i) State the objectives of and requirements for building a model. [6]

Helen believes that she can generate sufficient profits from betting on the results of sporting events to pay for her day to day living expenses.

She has identified that a model needs to be built to analyse past data and use this to predict future results.

- (ii) Describe the decisions that need to be made when determining the modelling approach to be taken to assess these future results. [6]

- (iii) Outline the characteristics of the cashflows which need to be incorporated into the model. [4]

[Total 16]

CA1-P1 | September 2017 | Q7

Current topic | 31 marks total | Original PDF p5

Related: Ch 7, Ch 27.

- 7** A professional orchestra that performs classical music concerts in a few locations around one city is about to undertake its first international tour. The objective of the tour is to raise the profile of the orchestra and generate income both now and into the future.

Over three weeks there will be a twelve concert tour across six different countries. There are about 150 people of many different nationalities in the tour entourage including musicians and support staff.

- (i) Outline the additional or extensions to existing insurance cover the orchestra might need for the tour over and above the cover it has in place for its regular domestic activities. [5]
- (ii) Discuss the factors the managers of the orchestra should consider when deciding whether to purchase additional insurance cover or to retain the additional risks. [6]

The tour director has asked an actuary to model its future cashflows to assess the tour's profitability and/or likelihood of achieving its objectives.

- (iii) State the basic features of general actuarial models that are used to project future cash flows.

You do not need to cover the steps or requirements for actually using or building a model. [5]

- (iv) Describe the cash flows relating to this tour that would need to be allowed for by a model that aims to assess the tour's profitability. [10]
- (v) Set out how sensitivity analysis would be used in conjunction with modelling the cash flows described in (iv). [5]

[Total 31]

CA1-P1 | September 2018 spare | Q2

[Current topic](#) | [8 marks total](#) | [Original PDF p2](#)

2018 spare paper (X18), distinct from the September main sitting. Original paper and matching spare examiner report restored from the supplied archive.

- 2** Describe the objectives of and requirements for building a model to project the profits of a global retail company. [8]

CP1-P1 | September 2020 | Q8

[Current topic](#) | [19 marks total](#) | [Original PDF p4](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 8** (i) Discuss the factors to be considered by an insurance company when deciding whether to use actual policy records or representative model points when using a financial model. [7]

An insurance company has recently completed a valuation of its liabilities for a particular class of business using its stochastic model. It has decided to review the accuracy of the results using actual policy records.

The company has already checked the accuracy of the data, assumptions and formulae to be used.

- (ii) Discuss the reasons why the insurance company has used a stochastic model rather than a deterministic model. [8]
- (iii) Set out the remaining steps that could be taken to review the accuracy of the model's calculated result. [4]

[Total 19]

CP1-P1 | September 2024 | Q5

Current topic | 7 marks total | Original PDF p2

Examiner report and original paper: use the links for this question in [index.html](#).

- 5** A general insurance company is reviewing its model that estimates the costs of Incurred But Not Reported (IBNR) claims and has commissioned a consultancy to undertake a review of the model and, if required, recommend changes to the model. The consultancy has carried out its review and has recommended that the insurance company replace its existing IBNR model, with an IBNR model that the consultancy had previously built for one of the insurer's competitors.

Discuss the issues the insurance company will need to consider before it decides to use the IBNR model that the consultancy had developed for another client. [7]